

What if one single entry point could route requests, balance traffic, secure APIs, and protect your entire microservices architecture?



What is an API?



- API = Application Programming Interface
- Allows different applications to communicate
- Acts as a bridge between client and server
- Sends requests and receives responses
- Example: Mobile App ↔ Backend API



What is an API Gateway?



- Single entry point for all client requests
- Sits between clients and microservices
- Receives requests and forwards them to the correct service
- Simplifies communication in distributed systems
- Commonly used in Microservices Architecture



What is Routing?



- Directs requests to the correct microservice
- Based on URL, path, or rules
- Example:
 - /auth → Auth Service
 - /orders → Order Service
 - /payments → Payment Service
- Eliminates direct client-to-service communication



What is Load Balancing?



- Distributes traffic across multiple service instances
- Prevents server overload
- Improves performance and availability
- Supports horizontal scaling
- Example:
 - Request 1 → Server A
 - Request 2 → Server B
 - Request 3 → Server C



What is Authentication (Auth)?



- Verifies user identity
- Ensures only authorized users access resources
- Common methods:
 - JWT Tokens
 - OAuth 2.0
 - API Keys
- First security layer of modern systems



What is Kong?



- Open-source API Gateway
- Designed for Microservices and Cloud-Native systems
- Handles routing, authentication, rate limiting, and load balancing
- Easy integration with Kubernetes and Docker
- Widely used in enterprise environments



API Gateway Architecture Flow



- Client sends request
- API Gateway receives request
- Authentication is verified
- Request is routed to correct service
- Service processes request
- Response returns through Gateway

